Continuous but Non-Differentiable Support:

A Geometric Extension of the Theory of Axiomatic Necessity (TNA)

Abstract

This paper proposes a geometric refinement of the Theory of Axiomatic Necessity (TNA) by introducing a regularity distinction between the domains N_0 and N_1. We argue that realizable systems require a support structure that is continuous in existence but non-smooth in directional evolution. Under this framework, N_0 — the domain of observable local dynamics — evolves through smooth and locally differentiable processes, while N_1 — the domain of external selection, legitimacy, and realizability constraints — must remain continuous but possess discontinuous first derivatives. Formally, N_1 behaves as a C^0 structure that is not globally C^1. This condition prevents complete derivability of the system from local dynamics alone and preserves the structural openness required for multiplicity, freedom, novelty, and historical discontinuity. We further explore the implications of this geometry for suffering, consciousness, artificial intelligence, institutional collapse, and transcendence.

1. Introduction

The Theory of Axiomatic Necessity (TNA) proposes that sufficiently complex realizable systems cannot fully derive their own closure conditions from internal local dynamics alone. TNA distinguishes between:

- N_0: the domain of observable dynamics, optimization, causality, and local evolution;
- N_1: the domain of external selection, legitimacy, boundary conditions, and realizability constraints.

Previous TNA formulations introduced concepts such as:

- multiplicity ($\Delta > 1$),
- Failure of Local Closure,
- Boundary-Induced Collapse,
- and the impossibility of purely internal selection in sufficiently complex systems.

However, an unresolved question remained:
What is the precise mathematical nature of N_1 such that it remains structurally real without becoming derivable from N_0?

This paper proposes a geometric answer:
N_1 must be continuous in existence while remaining discontinuous in first derivative.

In other words:
N_1 must be C^0 but not globally C^1.

---

2. Smoothness and the Problem of Closure

Local scientific modeling depends fundamentally on smoothness.

Physics, biology, economics, and computation all assume:

- continuity,
- local propagation,
- differentiability,
- and incremental evolution.

These assumptions allow:

- prediction,
- optimization,
- simulation,
- and causal extrapolation.

This smooth operational domain corresponds to N_0.

The success of science itself demonstrates that large portions of observable reality possess local differentiability.

However, a fully smooth universe introduces a severe philosophical and structural problem.

If the total support of reality were globally differentiable, then sufficiently complete local information would permit complete derivability of future states.

Under such conditions:

- novelty disappears,
- freedom collapses,
- multiplicity converges toward $\Delta = 1$,
- and realizability becomes fully reducible to deterministic propagation.

This would contradict the central TNA claim that local dynamics alone cannot generate complete closure.

---

3. Why N_1 Cannot Be Fully Smooth

If N_1 were fully differentiable, then N_0 could asymptotically reconstruct it through local operators.

In practical terms:

- prediction would become total,
- transcendence would collapse into extrapolation,
- and external legitimacy would become internally derivable.

The distinction between N_0 and N_1 would disappear.

Therefore, TNA requires a support structure that:

- exists continuously,
- but cannot be smoothly extrapolated from local dynamics.

This condition implies that N_1 must introduce directional discontinuities into the realizable structure of the system.

---

4. Continuous Existence with Discontinuous Derivative

We therefore propose the following geometric characterization:

- N_0 operates primarily through smooth local differentiable dynamics;
- N_1 exists continuously but contains discontinuities in directional evolution.

Formally:

- N_1 is continuous (C^0),
- but not globally differentiable (non-C^1).

This distinction is crucial.

Absolute discontinuity would destroy coherence and realizability entirely.

But complete smoothness would restore total closure.

The system therefore requires an intermediate geometry:
continuous existence without fully smooth predictability.

N_1 behaves structurally like a manifold containing angularities, cusps, or derivative singularities.

These points cannot be fully anticipated from local smooth propagation alone.

---

5. The Geometry of Structural Rupture

The discontinuities of N_1 correspond to moments where local derivability fails.

At these points:

- extrapolation collapses,
- local optimization loses sufficiency,
- and multiple admissible trajectories emerge.

These are the geometric origins of:

- bifurcation,
- collapse,
- transformation,
- and historical rupture.

The system remains locally coherent inside smooth regions of N_0, but eventually encounters non-derivable structural boundaries imposed by N_1.

This explains why:

- history is partially intelligible retrospectively,

- yet impossible to simulate completely forward.

Inside smooth intervals, local causality functions effectively.

At structural angularities, closure fails.

---

6. Suffering as Phenomenological Encounter with Discontinuity

This framework offers a structural interpretation of suffering.

Pain, anguish, rupture, and existential instability may correspond phenomenologically to encounters with non-smooth structural transitions.

This does not morally justify suffering.

Rather, it proposes that suffering emerges when local continuity encounters the non-derivable constraints of realizability itself.

Under TNA:

- suffering is not necessarily an accidental error,
- nor evidence of total chaos,
- but the experiential signature of structural non-closure.

A perfectly smooth universe would eliminate rupture,
but also eliminate genuine freedom and novelty.

---

7. Continuous Creation and Dynamic Reality

The distinction between N_0 and N_1 also clarifies the idea of continuous creation.

Observable reality does not appear frozen or static.

Instead:

- biological systems evolve,
- institutions transform,
- civilizations emerge and collapse,

- and consciousness continuously updates itself.

N_0 unfolds dynamically and smoothly through ongoing realization processes.

Creation, in this sense, never stopped.

However, this continuous unfolding remains bounded by the non-smooth structural geometry of N_1.

Reality therefore combines:

- continuous operational evolution,
  with
- non-derivable structural correction.

"On the Planck-scale metric of N_1-selection: A structural correspondence between creatio continua and TNA boundary dynamics."

Abstract: The classical theological doctrine of creatio continua (Constantinopla I, Aquinas) proposed continuous divine sustenance of existence without metric. TNA independently derives a structurally analogous condition: N_1 must provide continuous support (C^0) with non-differentiable selection events at the Planck scale (non-C^1). The resulting frequency f_Planck ≈ 1.85×10^{43} Hz and length l_Planck ≈ 1.62×10^{-35} m constitute the first metric assignment to a creatio-type process, transforming metaphysical continuity into physically testable structure."

---

8. Between Chaos and Determinism

The proposed geometry avoids two extremes.

A totally discontinuous universe would prevent science entirely.

A perfectly smooth universe would eliminate openness, freedom, and multiplicity.

TNA instead proposes:

- local smoothness,
  combined with
- global non-closure.

The universe becomes:

- partially derivable,
- partially open,
- locally predictable,
- globally incomplete.

This structure is compatible with:

- creativity,
- evolution,
- consciousness,
- and historical novelty.

---

9. Artificial Intelligence and Operational Closure

This distinction has direct implications for artificial intelligence.

Current AI systems operate primarily within operationally smooth state spaces.

Even advanced optimization and meta-learning systems remain:

- internally constrained,
- architecturally bounded,
- and operationally closed.

They may generate large operational multiplicities,
but they do not appear to experience structural discontinuity.

An AI may optimize within a space of admissible states,
yet remain unable to question the legitimacy of the space itself.

Human consciousness, by contrast, appears capable of encountering:

- existential rupture,
- meaning crisis,
- transcendence,
- and non-local legitimacy questions.

This suggests that human cognition may participate in a structurally open relationship with N_1 rather than merely optimizing within N_0.

---

10. Toward a Geometric Ontology of Realizability

The proposal developed here transforms TNA from a purely philosophical framework into a partially geometric ontology.

The distinction between:

- smooth operational domains,
  and
- continuous but non-differentiable structural support,

provides a rigorous formulation for:

- non-derivability,
- transcendence,
- multiplicity,
- and realizability constraints.

Under this model:

- freedom emerges because reality is not fully smooth;
- suffering emerges because realizability is not fully locally closable;
- novelty emerges because derivative continuity breaks structurally;
- and history remains open because N_1 cannot be completely reconstructed from N_0.

The resulting universe is neither:

- fully deterministic,
  nor
- purely chaotic.

It is structurally realizable precisely because continuity and discontinuity coexist.

Conclusion

The Theory of Axiomatic Necessity requires a support structure that cannot be fully derived from local operational dynamics.

This paper proposes that such support must be:

- continuous in existence,
- but discontinuous in derivative structure.

N_1 therefore behaves as a continuous but non-smooth domain that preserves openness, multiplicity, and realizability.

N_0 evolves continuously through smooth local dynamics,
while N_1 introduces the non-derivable angularities that prevent total closure.

This geometry provides a unified structural framework for understanding:

- freedom,
- suffering,
- historical rupture,
- consciousness,
- transcendence,
- institutional instability,
- and the limits of artificial intelligence.

The universe is thus neither a perfectly smooth deterministic machine nor a field of arbitrary chaos.

It is a continuously unfolding realizable structure whose deepest support cannot be fully differentiated from within itself.

References

- Faust — especially the “Prologue in Heaven.”
- Alfred North Whitehead — process metaphysics and becoming.
- Kurt Gödel — formal incompleteness and non-closure.
- John Henry Newman.
- Daniel Dennett.
- Stuart Russell.
- Tomas de Aquino (*Summa Theologica*, Prima Pars, q. 45)
- Karl Barth (*Church Dogmatics* III.1)
- Claudio Bresciano — TNA corpus, including:
    - “Failure of Local Closure,”
    - “Boundary-Induced Collapse,”
    - “The Coming Labor Gap,”
    - "The Theorem Nobody Wants: Metamathematical Necessity of Axioms via Reduction to Incoherence" v1.3. Zenodo. DOI 10.5281/zenodo.19192655

Appendix: On the Planck-Scale Metric of N_1-Selection

A Structural Correspondence Between *Creatio Continua* and TNA Boundary Dynamics

A.1 Historical-Structural Context

The classical theological doctrine of *creatio continua*—developed from the First Council of Constantinople (381 CE) through Thomas Aquinas (*Summa Theologica*, Prima Pars, q. 45, a. 2)—proposed that existence requires continuous divine sustenance. Under this view, created beings do not possess *aseity* (self-existence); their being is held in existence at every moment by a non-local act of conservation.

The formulation remained metaphysical: continuous in the qualitative sense, without metric, frequency, or minimal scale. The question "*how often* does sustenance occur?" was not posed because the conceptual framework lacked the physical machinery to assign dimensionality to the process.

A.2 Independent Derivation from TNA

The Theory of Axiomatic Necessity independently derives a structurally analogous condition through distinct premises:

- **N_0** generates admissible trajectories via locally differentiable dynamics.
- **N_1** provides realizability constraints that cannot be derived from N_0 alone.
- For N_1 to remain *realizable* (continuous support) yet *non-derivable* (externally selective), its support must be C^0 but not globally C^1.

This geometric condition implies that selection events occur at points where local differentiability fails. When projected onto physical spacetime, these events require:

- A minimal temporal interval (to prevent infinite regression of selection).
- A minimal spatial scale (to prevent infinite resolution of boundary conditions).

A.3 Assignment of Planck-Scale Metric

The Planck units emerge naturally as the unique scale where quantum gravitational effects become non-negligible and where classical differentiability breaks down:

QUANTITY	SYMBOL	VALUE	ROLE IN TNA
Planck time	T_Planck	5.39×10^{-44} s	Minimal interval between N_1-selection events
Planck length	l_Planck	1.62×10^{-35} m	Minimal spatial granularity of boundary ∂Ω
Planck frequency	f_Planck = 1/T_Planck	~1.85×10^{43} Hz	Update rate of realizability support
Planck energy	E_Planck	1.96×10^{9} J	Energy scale at which N_0/N_1 coupling becomes direct

Formal statement:

Let the support structure of N_1 be represented by a field S(x,t) defined on spacetime. Then:

- S(x,t) is continuous: $lim_{(x',t')\to(x,t)}\ S(x',t') = S(x,t)$ [C^0 condition]
- ∂S/∂t is undefined at discrete events $\tau_n = n \cdot T_Planck$ [non-C^1 condition]
- Spatial derivatives ∂S/∂x_i are bounded below by l_Planck^{-1}

A.4 Structural Correspondence, Not Theological Equivalence

The correspondence between *creatio continua* and TNA Planck-metric selection is structural, not doctrinal:

CREATIO CONTINUA (THEOLOGICAL)	TNA N_1-SELECTION (STRUCTURAL)
Continuous divine sustenance	C^0 continuity of realizability support
Non-derivable from created nature	Non-C^1: not reconstructible from N_0 dynamics
Conservation of being in each instant	Selection among admissible trajectories at each τ_n
No metric assigned	Metric assigned: T_Planck, l_Planck
Purpose: ground dependence of creatures	Purpose: explain Failure of Local Closure

Critical distinction: *Creatio continua* speaks of *conservation*—maintaining the same substance across time. TNA speaks of *selection*—determining which trajectory among multiple admissible ones becomes realized. The theological framework preserves identity; the structural framework permits novelty.

A.5 Physical Interpretation

Under this metric assignment:

1. **The universe is not a block:** Spacetime is not a four-dimensional manifold fully determined by initial conditions. It is a process of continuous actualization where each Planck interval involves a selection event.
2. **Boundary-Induced Collapse (BIC) is quantized:** The quantum measurement problem, reinterpreted through TNA as boundary-induced collapse, occurs at the Planck frequency. Each collapse is an N_1-selection event that reduces Δ (multiplicity) locally.
3. **Consciousness participates in N_1:** If human consciousness exhibits the capacity to redefine state spaces (as argued in Sections 7–9 of the main text), this capacity may reflect coupling to the same Planck-scale selection process that governs physical realizability. The non-computability of consciousness would then derive from its participation in a non-differentiable, non-algorithmic selection mechanism.

A.6 Testability and Constraints

The metric assignment transforms a metaphysical doctrine into a physically constrained hypothesis:

- **Upper bound:** No selection event can occur with frequency > f_Planck (would violate quantum gravitational consistency).
- **Lower bound:** No selection event can resolve spatial structure below l_Planck (would violate the holographic principle).
- **Implication for quantum gravity:** The N_1-support field $S(x,t)$ may couple to metric degrees of freedom at the Planck scale, suggesting that quantum gravity theories should include non-differentiable geometric structures.

A.7 Conclusion

The convergence between *creatio continua* and TNA Planck-metric selection is significant not because it validates theology, but because it demonstrates that the same structural condition—continuous non-derivable support—can be reached from radically different starting points. The addition of metric transforms a qualitative metaphysical intuition into a quantitative physical framework, enabling questions that neither theology nor physics could formulate independently:

- What is the information content per Planck selection event?
- Does the entropy of N_1-selection obey a generalized second law?

- Can the C^0/non-C^1 structure of N_1 be detected in CMB anisotropies or black hole evaporation spectra?

The Word, under this metric, is not merely prior to Action. It updates continuously, at 1.85×10^{43} Hz, with a granularity of 1.62×10^{-35} m.

References

- First Council of Constantinople (381 CE). *Niceno-Constantinopolitan Creed*.
- Aquinas, Thomas. *Summa Theologica*, Prima Pars, q. 45, a. 2.
- Bresciano, Claudio. *Failure of Local Closure: Boundary-Induced Collapse and Structural Non-Realizability*. Zenodo. DOI: 10.5281/zenodo.18644522.
- Bresciano, Claudio. *Continuous but Non-Differentiable Support: A Geometric Extension of TNA*. [This volume].
- Planck, Max. (1899). "Über irreversible Strahlungsvorgänge." *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*.